Yashaj Gupta

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Education

- **Jabalpur Engineering College** — Jabalpur, India
  B.Tech in Electronics and Telecommunication Engineering; GPA: 6.5 (Till Sem 6) — *2023 – 2027 (Expected)*

Skills

- **Programming**: C, C++, Embedded C, Python, SQL
- **Frameworks & Libraries**: Qt (Widgets/QML), OpenCV, TensorFlow
- **Embedded Platforms**: Arduino, STM32, ESP32, Raspberry Pi
- **Tools**: Git, CMake, MySQL, SQLite, Qt-SQL
- **Concepts**: Data Structures & Algorithms, Embedded Systems, IoT, Software Development
- **Soft Skills**: Leadership, Technical Writing, Public Speaking, Time Management

Experience

- **Noctilucent Technology** — On-site
  Embedded Systems Intern — *Dec 2025 – Feb 2026*
  - **Industrial System Development**: Built a hazard detection and training board for deployment at a **Reckitt** factory (Rudrapur).
  - **Sensor Integration**: Designed and integrated sensor-based detection system for identifying unsafe conditions in real-time.
  - **Firmware Development**: Programmed microcontroller-based firmware in C/C++ for monitoring and alert generation.
  - **System Testing & Debugging**: Performed hardware and firmware debugging to ensure stable and accurate operation in real-world environments.
- **Gradnary** — Remote
  Founder's Associate Intern — *Dec 2024 – Feb 2025*
  - **SEO & Web Optimization**: Enhanced digital visibility using Google Analytics and keyword research.
  - **Data Automation**: Developed Python + Excel scripts to organize structured datasets for reporting.
  - **Project Coordination**: Built internal knowledge base and managed tasks via Trello and Notion.

Projects

- **Smart Attendance System (Campus Deployment)**: **Stack:** ESP32, Raspberry Pi, Wi-Fi, Database — Built a building-wide attendance system with ESP32 nodes and a Raspberry Pi server maintaining a centralized database with restricted teacher-only access and real-time data synchronization.
- **Smart Canteen QR Payment System**: **Stack:** ESP32, C++, Arduino, Firebase/SQL, LCD, Keypad, QR Scanner — Designed a real-time wallet system where student ID cards act as payment tokens with secure QR + PIN-based transactions.
- **Colour Detection System**: **Stack:** Python, OpenCV — Implemented real-time color detection using HSV filters and masking for robotics vision applications.
- **SIH Rover Project – Autonomous Agricultural Rover**: **Stack:** ROS2, C++, OpenCV, Arduino, STM32, Gazebo — Designed and simulated a semi-autonomous rover for Smart India Hackathon (SIH) with precision pesticide spraying, terrain navigation, and vision-based obstacle detection.
- **E-Yantra Self-Balancing Drone – IIT Bombay**: **Stack:** Raspberry Pi 4, Python, PID Control, Image Processing, IMU Sensors — Built a self-balancing quadcopter with flight path recording and sensor fusion, achieving stable hover and autonomous flight.

Research Work

- **ABS (Anti-lock Braking System) Simulation**: **Stack:** MATLAB Simulink — Proposed an adaptive ABS model with slip-ratio control and dynamic hydraulic pressure distribution to enhance braking stability and safety.

Certificates

- **50 Days to C++ : From Zero to Pro Developer (Udemy, 2025)**
- **Introduction to C++ : Coding Ninjas (2024)**
- **Qt 6 Core with C++ : Udemy (2025)**

Volunteer Experience

- **Training and Placement Office, JEC** — India
  Assisted in managing and planning campus placement events — *Dec 2023 – Present*
- **IoT and Robotics Lab, JEC** — India
  Assisted in conducting IoT and robotics experiments and student project demonstrations — *July 2025 – Present*